

MONTREAL-EST, QC, Canada

WMO: 716751

Lat: 45.630N

Lon: 73.550W

Elev: 50

StdP: 100.72

Time Zone: -5.00 (NAE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -20.2	(d) -30.0	(e) 0.2	(f) -23.1	(g) -27.3	(h) 0.3	(i) -20.4	(j) 10.2	(k) -0.7	(l) 9.2	(m) -2.8	(n) 3.8	(o) 260	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 30.5	(d) 21.0	(e) 29.0	(f) 20.1	(g) 27.6	(h) 19.4	(i) 22.7	(j) 27.7	(k) 21.7	(l) 26.2	(m) 20.9	(n) 25.1	(o) 4.4	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	15.9	24.8	20.2	15.0	24.3	19.3	14.2	23.7	67.4	28.0	63.7	26.0	60.6	24.9	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.6	(b) 7.6	(c) 6.8	(e) -25.1	(f) 32.6	(g) 3.1	(h) 1.4	(i) -27.3	(j) 33.6	(k) -29.1	(l) 34.4	(m) -30.8	(n) 35.3	(o) -33.0	(p) 36.3		
(4) 8.6	(5) 7.6	(6) 6.8	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-8.8	-7.3	-1.6	6.2	13.8	19.4	21.4	20.7	16.4	9.2	2.4	-4.5
	DBStd	11.71	6.99	5.91	5.74	4.66	4.20	3.95	2.82	3.11	3.96	4.50	4.80	5.73
	HDD10.0	2321	583	484	359	129	11	0	0	0	2	71	233	449
	HDD18.3	4294	842	717	617	363	152	34	6	12	83	283	478	707
	CDD10.0	1357	0	0	1	17	128	283	353	332	193	47	4	0
	CDD18.3	289	0	0	0	1	11	67	100	86	24	2	0	0
	CDH23.3	2057	0	0	0	6	96	535	683	583	148	6	0	0
	CDH26.7	502	0	0	0	1	19	151	158	139	34	0	0	0
Wind	WSAvg	3.4	3.6	3.5	3.6	3.6	3.5	3.2	3.1	2.9	3.0	3.4	3.5	3.5
Precipitation	PrecAvg	1029	73	60	65	87	90	94	97	95	93	102	82	92
	PrecMax	1303	147	110	123	158	173	163	152	195	190	188	135	155
	PrecMin	810	27	25	20	18	43	41	53	38	29	32	16	29
	PrecStd	123	32	21	24	39	32	29	24	35	37	40	28	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	6.8	14.5	23.7	29.3	32.1	31.7	31.9	30.5	24.1	17.2	11.1
		MCWB	4.8	4.5	9.4	15.1	17.7	21.1	23.3	21.6	20.2	18.0	14.5	9.7
	2%	DB	5.9	4.8	10.6	19.5	26.0	30.2	29.8	29.7	27.0	21.1	14.3	6.9
		MCWB	3.8	2.8	6.1	11.1	15.0	20.1	21.6	20.6	19.5	15.4	11.9	5.3
	5%	DB	3.3	3.0	7.8	16.2	23.4	28.4	28.4	28.1	24.7	18.6	11.5	4.7
		MCWB	1.7	1.6	4.4	9.2	14.3	19.4	20.2	19.8	18.3	14.0	9.2	3.3
	10%	DB	1.0	1.5	5.9	13.4	21.2	26.6	27.0	26.6	22.7	16.1	8.9	2.9
		MCWB	0.2	0.0	3.4	7.6	13.7	18.2	19.6	19.2	17.3	12.4	6.5	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	5.7	10.5	16.3	19.7	23.1	24.6	23.4	21.9	18.7	14.7	10.0
		MCDB	7.9	6.6	13.0	20.0	24.9	28.0	29.8	29.5	25.6	22.8	16.6	11.6
	2%	WB	2.6	3.6	7.0	12.8	17.9	21.6	23.2	22.3	20.5	16.2	12.1	5.6
		MCDB	4.1	4.8	9.1	17.4	22.5	26.4	28.1	27.5	25.2	19.7	14.1	7.0
	5%	WB	1.1	1.9	5.1	10.2	16.5	20.6	22.1	21.4	19.3	14.2	9.4	3.5
		MCDB	2.0	3.1	7.6	14.6	21.5	25.2	26.3	25.7	23.6	18.0	11.5	4.7
	10%	WB	-1.0	0.2	3.5	8.1	15.0	19.5	21.0	20.5	18.1	12.5	6.8	1.8
		MCDB	0.2	1.5	5.7	12.2	19.5	24.2	25.0	24.7	21.7	15.4	8.6	2.8
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	7.7	8.8	9.5	9.2	8.5	8.9	8.8	7.3	6.0	6.2
		MCDBR	9.4	8.3	10.0	13.4	12.7	11.4	10.0	10.3	10.7	10.4	9.6	7.3
		MCWBR	10.8	7.8	7.3	7.9	5.7	5.0	4.6	4.5	5.4	6.4	8.0	7.0
	5% WB	MCDBR	9.4	8.6	10.1	11.8	10.5	9.2	8.5	9.1	9.1	10.2	9.2	7.3
		MCWBR	9.2	8.1	8.1	8.2	6.3	5.4	4.9	5.2	5.5	7.0	8.2	6.9
Clear-Sky Solar Irradiance	taub	0.257	0.289	0.334	0.363	0.390	0.439	0.428	0.414	0.374	0.336	0.307	0.293	
	taud	2.349	2.250	2.263	2.353	2.339	2.206	2.244	2.280	2.380	2.492	2.487	2.299	
	Ebn at Noon	863	888	891	896	878	832	837	834	841	831	797	759	
	Edh at Noon	74	100	116	116	122	140	133	124	103	80	67	71	
All-Sky Solar Radiation	RadAvg	1.44	2.38	3.63	4.38	5.12	5.63	5.64	4.99	3.94	2.28	1.38	0.99	
	RadStd	0.17	0.25	0.30	0.50	0.49	0.42	0.43	0.23	0.32	0.20	0.18	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page